

Patterns of Life

KEY QUESTIONS

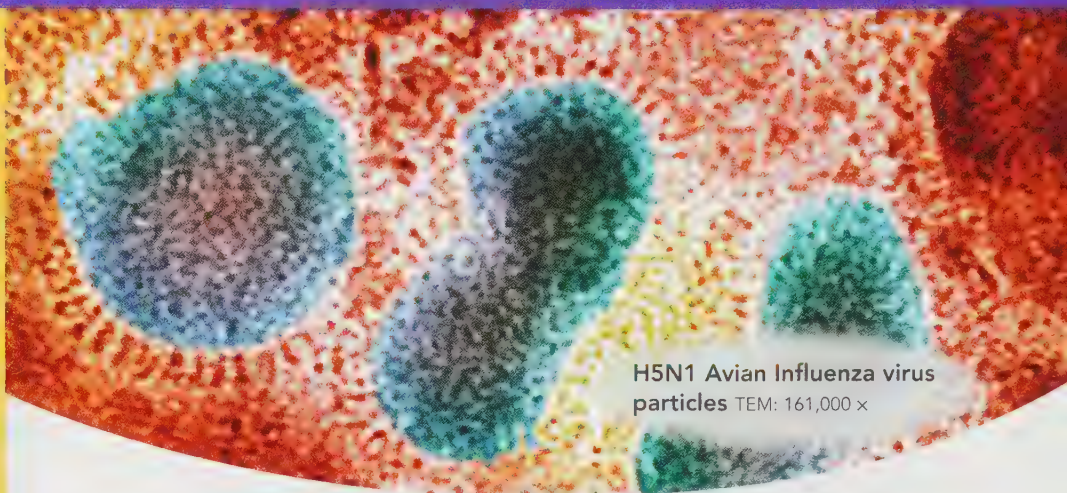
- What characteristics do all living things share?
- What are the crosscutting concepts of biology?
- How do fields of biology differ in their approaches?
- How is the metric system important to science?

VOCABULARY

biology
sexual reproduction
asexual reproduction
DNA
metabolism
stimulus
homeostasis
evolve

READING TOOL

Complete the chart in your **Biology Foundations Workbook** by writing the main idea and details from this lesson on the characteristics of life.



H5N1 Avian Influenza virus particles TEM: 161,000 ×

Think about some of the important and exciting news stories you've read or heard about recently. Bird flu spreads around the world, killing thousands of birds and threatening a human epidemic. Users of certain illegal drugs experience permanent damage to their brains and other parts of their nervous systems. Reports surface about the use of cloned human cells to grow new organs to replace those lost to disease or injury. These and many other stories involve **biology**—the science that employs scientific methodology to study living things. (The Greek word *bios* means "life," and *-logy* means "study of.")

Characteristics of Living Things

Biology is the study of life. But what is life? What distinguishes living things from nonliving matter? Surprisingly, it isn't as simple as you might think to describe what makes something alive. No single characteristic is enough to describe a living thing. Also, some nonliving things share one or more traits with organisms. For example, a firefly and a fire both give off light, and each moves in its own way. Mechanical toys, automobiles, and clouds (none of which are alive) move around, while mushrooms and trees (which are alive) stay in one spot. To make matters more complicated, some things, such as viruses, exist at the border between living and nonliving things.

Despite these difficulties, we can list characteristics that most living things have in common. **Living things are made up of basic units called cells, reproduce, are based on a universal genetic code, grow and develop, obtain and use materials and energy, respond to their environment, maintain a stable internal environment, and evolve, changing over time.**